

## Skills

- **Programming Languages** : Javascript, TypeScript, Java, Python, HTML, CSS, C++
- **Libraries and Frameworks** : React, React Native, Redux, Node.js, Spring Boot, Next.JS, Angular, Fast API, Flask, Django, Express.js
- **Tools** : MySQL, PostgreSQL, AWS, Kubernetes, GCP, MongoDB, Docker, Microservices, Git, CI/CD

## Experience

### Forward Deployed Engineer

Jun 2022 - Present

#### Skillnet Solutions Inc

- Provisioned a centralized devbox on AWS EC2 with Tailscale for secure remote access to all hosted services, and authored Terraform-based infrastructure automation using EventBridge and Lambda to auto-shutdown idle servers on a schedule, cutting monthly cloud costs by 15%.
- Architected platform-agnostic microservices with standardized REST APIs across VTEX, Spryker, SFCC, Medusa, kibo & Shopify — partnering directly with client engineering teams to enable plug-and-play integration, reducing delivery speed by 40%.
- Developed high-throughput backend services and asynchronous job processing pipelines using RabbitMQ (FastApi), optimizing server-side data handling and payload compression to deliver throughput improvement and a 20% increase in sales conversion.
- Independently designed and shipped end-to-end e-commerce features — bonus-point deduction at checkout with non-refundable balance enforcement, infinite scroll product listing, review & rating system, and gift card issuance/redemption — from API design through production rollout.
- Built server-side routing infrastructure using middleware, enforcing canonical URL resolution, HTTP 301 redirect logic, slug-based routing, and JSON-LD schema markup injection — improving organic search rankings by 12 positions and crawl efficiency by 45%.
- Architected a shared internal service layer and SDK encompassing product catalog, cart, checkout, and pricing domains using versioned REST and GraphQL APIs in Node.js, eliminating duplicate business logic across squads and accelerating feature delivery by 40%.
- Diagnosed and resolved missing GA4 order-tracking data caused by caching before tag fired cross-verified records via DBeaver and restored accurate order visibility and Google Ads/Meta conversion tracking.

## Projects

### WILCO FARMSTORE

🔗 <https://www.farmstore.com/>

- Re-architected monorepo tooling by replacing Lerna with Turborepo, consolidating into a single command, and migrated repository infrastructure from Bitbucket to GitLab for seamless transition.
- Engineered and optimized a centralized reporting module that ingested and normalized FTP/EDI data, automating export workflows to deliver actionable insights on inventory and store-level performance.
- Designed a robust inventory management system to seamlessly reconcile stock levels, product variants, and location-specific data, utilizing strict validation pipelines to ensure cross-channel inventory accuracy.
- Developed a comprehensive vendor management system to standardize supplier data and track complex product mappings, streamlining downstream analytics and empowering data-driven decisions.

**TechStack:** Angular, PostgreSQL, MySQL, Redis, Kubernetes, GraphQL, Tailwind CSS, ArgoCD, Terraform, Helm, Sentry, Google Analytics, UNBXD, Back40, RabbitMQ, TypeScript

### Follett Higher Education

🔗 <https://follett.com/>

- Architected a microservices-based e-commerce platform (Login, Authorization, Product Catalog, Payment Gateway, Service Discovery, Notification) to replace a monolithic setup, enabling independent scaling/deployment and isolating failures to individual services.

- Deployed the ecosystem on AWS Elastic Beanstalk with RDS and CloudWatch, cutting deployment overhead and reducing mean time to detect issues by 30%.
- Built an event-driven email service using Kafka to decouple notifications from core services, cutting checkout/order-flow latency by 30% at scale.
- Implemented sorting, filtering, and pagination for product discovery, reducing average search/browse response time by 30%.
- **Tools:** Spring Boot, Spring Cloud, MySQL, Hibernate, Redis, Razorpay, Stripe, JUnit, Kafka

## RAG PIPELINE

🔗 <https://github.com/aamirbohra72/RAG-Pipeline>

- Built a production-ready document Q&A platform using FastAPI, Next.js, LangGraph enabling users to upload documents and receive streaming, context-grounded answers with source citations.
- Implemented an asynchronous document ingestion pipeline using Celery and Redis, supporting PDF, Markdown, HTML, CSV, and XLSX processing with OCR fallback through PyMuPDF, RapidOCR .
- Engineered hybrid retrieval using pgvector, ChromaDB, embeddings, lexical similarity, and cross-encoder reranking, with user-scoped document isolation and relevance filtering to improve answer quality.
- Developed conversational RAG workflows with LangGraph — query rewriting, multi-turn history, and LangSmith tracing for observable latency and token-usage metrics.

## Yootube Joy

🔗 <https://play.google.com/store/apps/details?id=com.youtubeproject>

- Engineered a React Native parental control app using Redux Toolkit with role-based access control (RBAC) enabling parents to filter and manage children's content consumption across 5,000+ active users.
- Built secure authentication APIs using Node.js, Express, and Bcrypt with OTP-based verification via Brevo, achieving 99.8% email delivery rate and reducing account creation friction by 30%.
- Integrated Razorpay payment gateway and subscription management with a 3-day free trial, converting 40% of trial users to paid plans and generating ₹2L+ monthly recurring revenue.
- Implemented dynamic content filtering using YouTube API integration, allowing real-time playlist curation and reducing inappropriate content exposure by 95%.

## Competitive Programming Platform

🔗 <https://github.com/aamirbohra72/Competitive-Programming-Platform>

- Architected a full-stack competitive programming judge with a PostgreSQL and Redis data layer, delivering real-time verdict scoring, cached leaderboards, and cutting manual grading from hours to seconds.
- Engineered an AI interview chatbot (utilizing mic/cam, speech-to-text) to auto-generate candidate skill-gap reports, paired with a dynamic resume builder that compiles user metrics into professional portfolios.
- Built a AI-style course creator featuring MCQs and automated certificates, supported by an integrated AI TA chatbot for instant doubt resolution and a Visual Algo tool for step-by-step algorithm tracing.

## DiagramAI

🔗 <https://draw-app-web.onrender.com>

- Built two Mistral AI-powered pipelines — text-to-diagram and Mermaid-to-diagram — that convert natural language descriptions and flowchart syntax directly into an editable live canvas.
- Engineered real-time multi-user collaboration using WebSockets, TypeScript, and Node.js, enabling instant synchronized updates across sessions so multiple users can co-edit diagrams simultaneously.
- Developed a full-stack architecture integrating LLM-based parsing to transform unstructured natural-language input into structured, editable diagram objects, bridging free-form ideation with visual output.

## Achievements

- Won hackathon by building an AI-powered Virtual Try-On feature using Hugging Face and Mediapipe Pose Estimation, enabling real-time garment fitting with 85%+ accuracy on user-uploaded images.

## Education

**Maulana Azad National Institute of Technology, Bhopal**  
ME/M.Tech

2022